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1.0 **DOCUMENT CONTROL**

1.1 Distribution

Course Title	Advanced Certificate in Deep Learning and Generative AI
Document Title	Course Outline

1.2 Change Records

Version	Date	Record(s)	Author(s)
1.0	19 Apr 2024	Release Document	Ian Choo, Lui Huiliang, Chung Yong Yang
2.0	27 Jan 2025	Release Document	Ian Choo, Lui Huiliang

1.3 Content

1.0	Document Control
2.0	Course Outline
3.0	Module 1 – Introduction to Deep Learning
4.0	Module 2 – Deep Learning Building Blocks
5.0	Module 3 – Modern AI for Image and Natural Language Processing
6.0	Module 4 – Transformers, Large Language Models and Multimodal Learning
7.0	Module 5 – Deep Learning in Production
8.0	Module 6 – Tuning Deep Learning and Generative AI for Custom Applications

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2.0 COURSE OUTLINE

2.1 Certificate Title

Advanced Certificate in Deep Learning and Generative AI

2.2 Course Overview

This comprehensive 102-hour course equips you with the skills to build real world solutions to solve a vast array of business challenges with Generative AI (Gen AI) and Deep Learning, using tools and frameworks available in the Python ecosystem. Both prior knowledge of deep learning, and experience in Python programming, are not required.

The curriculum is structured into six modules, each spanning 17 hours. The journey begins with an introduction to Python Programming for Deep Learning and Generative AI (Gen AI), followed by an exploration of modern models for machine vision, text processing and handling various other forms of media and unstructured data. We then delve into the world of Large Language Models (LLMs) and Generative AI, including practical implementations of these models using various Python frameworks for Deep Learning like PyTorch and Tensorflow. The course concludes with the examination of real-world business applications of Deep Learning and Gen AI, as well as practical concerns around their deployment and optimisation.

2.3 Module Titles

- Module 1: Introduction to Deep Learning
- Module 2: Deep Learning Building Blocks
- Module 3: Modern AI for Image and Natural Language Processing
- Module 4: Transformers, Large Language Models (LLMs) and Multimodal Learning
- Module 5: Deep Learning in Production
- Module 6: Tuning Deep Learning and Generative AI for Custom Applications

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3.0 **Module OUTLINE**

3.1 **Module Title**

Introduction to Deep Learning

3.2 **Module Overview**

This module provides a foundational introduction to Deep Learning, designed for professionals with no prior technical background. Participants will develop essential Python programming skills and learn key mathematical concepts like vectors and tensors to understand neural networks. The broader landscape of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) is explored, with a focus on real-world applications across industries. Hands-on exercises using Python and no-code platforms allow participants to build and interact with basic models, while sessions on APIs and third-party tools demonstrate practical AI integration. This module equips learners with the groundwork for advanced topics in later modules.

3.3 **Module Objectives**

- Learning Objectives
  - Explore the core concepts of deep learning, and applications of deep learning in the real world
  - Understand vectors, spaces, and dimensionality as mathematical foundations
  - Describe how to setup a PyTorch or Tensorflow environment, to define and train a simple neural network
  - Interact with Deep Learning and Generative AI models via APIs and apps, using Python to access third-party providers like OpenAI, Anthropic, AWS Bedrock, and Google

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3.4 **Module Flow**

Session	Module Flow
SESSION 1 (2.5 HOURS)	**<u>Python Basics – Variables, Data Types, and Control Flow</u>** **Description** Introduce Python programming essentials, including variables, data types (strings, integers, lists, dictionaries), and control flow structures (if-else statements, loops). Practical exercises include basic arithmetic operations, iterating over lists, and implementing decision-making logic. This ensures learners can write simple programs and understand Python's core syntax. **Weightage of theory/hands-on for session:** • Theory: 75 mins • Hands-on: 75min
SESSION 2 (2.5 HOURS)	**<u>Python Functions and Data Structures</u>** **Description:** Learn to define and use functions for modular programming and explore Python's data structures, such as lists, tuples, sets, and dictionaries. Practical examples involve manipulating datasets, implementing reusable code blocks, and organizing structured data. Exercises reinforce learners' ability to manage and preprocess data effectively. **Weightage of theory/hands-on for session:** • Theory: 75 mins • Hands-on: 75 mins
SESSION 3 (3 HOURS)	**<u>Python Frameworks for Deep Learning</u>** **Description:** Dive into Python libraries used in Deep Learning workflows: ◦ **TensorFlow**: Understand tensors, build computation graphs, and implement simple models. ◦ **PyTorch**: Explore its dynamic computation graph and syntax by creating a basic PyTorch model. ◦ **Matplotlib**: Learn to visualize datasets, model performance, and results with plots. Hands-on exercises integrate these libraries into a cohesive Deep Learning workflow • Theory: 90 mins • Hands-on: 90 mins
SESSION 4 (1 HOURS)	**<u>Vectors, Spaces, and Dimensionality</u>** **Description:**

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	Introduce the concepts of vectors, vector spaces, and dimensionality. Explain how these concepts form the basis for tensors and embeddings in Deep Learning. Discuss applications such as representing text and images as vectors in high-dimensional spaces. Visualise relationships between vectors using simple diagrams to reinforce understanding. **Weightage of theory/hands-on for session:** • Theory: 40 mins • Hands-on: 20 mins
SESSION 5 (1.5 HOURS)	**Neural Networks Basics** **Description:** Cover the anatomy of a neural network, including neurons, layers, weights, biases, and activation functions. Discuss forward propagation and how neural networks learn by adjusting weights through backpropagation. Provide intuitive, high-level examples without delving into implementation yet. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 30 mins
SESSION 6 (4 HOURS)	**Overview of AI, Machine Learning, and Deep Learning** **Description:** Explore the evolution of AI, ML, and DL, focusing on their differences and overlaps. Highlight real-world applications like computer vision, fraud detection, and autonomous vehicles. Discuss the role of neural networks in advancing AI and provide historical context on breakthroughs like backpropagation and deep architectures. **Weightage of theory/hands-on for session:** • Theory: 160 mins • Hands-on: 80 mins
SESSION 7 (1.5 HOURS)	**Interacting with Deep Learning Models Through APIs Using Python** **Description:** Explore the landscape of third-party AI service providers, including **OpenAI**, **Anthropic**, **AWS Bedrock**, and **Google Vertex AI**. Compare their capabilities, such as text generation, summarization, and translation, and introduce their Python SDKs and API documentation. **Weightage of theory/hands-on for session:** • Theory: 45 mins • Hands-on: 45 mins
SESSION 5 (1 HOUR)	**Review and Assessment** **Description:**

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	The assessment will comprise a set of multiple-choice questions to reinforce the concepts covered in the module, as well as a practical component to test the participants on their ability to access third-party Deep Learning models via API with Python.

3.5 **Who Should Attend**

- Software Developers seeking to enhance their skillset with Deep Learning expertise
- Data Scientists looking to expand their toolbox with powerful Deep Learning techniques
- Researchers and Engineers interested in applying Deep Learning to their specific domains
- Anyone with a passion for artificial intelligence and a desire to build intelligent systems

3.6 **Assessment**

3.6.1 Type of Assessment / Course Work

3.6.2 A) State the mode of assessment and provide a breakdown of the assessment duration in each mode of assessment.

Multiple-Choice Questions (Individual, open-book) – 20 min
Practical Component (Individual, open-book) – 40 min

3.6.3 B) Write up on why the assessment method is appropriate for the target audience/context/ application of the course and the assessment duration is reasonable.

The Multiple-Choice Questions are open-book as participants are not expected to memorise any information, but be able to search and test for the relevant information to arrive at the correct answers. Thus, ample time of more than 1 min per question is given.

The Practical Component is open-book as well, since data scientists and machine learning engineers and practitioners have the freedom to look up and reference any information they require to achieve their tasks.

3.6.4 C) Write up/ Assessment Plan on how the assessment can help to assess the knowledge gained from the training and demonstrate how the learning outcome can be met.

The Multiple-Choice portion is a straightforward and objective way to assess participants' grasp of theoretical knowledge covered during the module.
The Practical Component assesses how well the participants are able to apply the skills that were covered in the module to a real-world example.

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3.7 **Certification**

3.8 **Classroom Type**

3.9 **Special instructions for participants**

3.10 **Instructor's Bio and Head Shot (High Res Professional photo)**
Note: Please attach the photo in email as well.

3.11 **Supporting documents for verification process**

3.11.1 **Credentials mentioned in Bios.**

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4.0 Module OUTLINE

4.1 Module Title

Deep Learning Building Blocks

4.2 Module Overview

This module delves into the core components of Deep Learning, providing participants with a practical understanding of the techniques and tools that underpin modern AI systems. Key topics include feedforward neural networks, loss functions, backpropagation, and regularisation methods to improve model generalisation. Participants will also learn hyperparameter tuning and transfer learning, using pre-trained models from the Hugging Face Model Hub to solve specific tasks efficiently. Practical exercises allow learners to apply these concepts to real-world scenarios, such as classification and prediction problems. By the end of this module, participants will have a solid grasp of the foundational elements of Deep Learning, preparing them to tackle advanced methods and applications.

4.3 Module Objectives

- Learning Objectives
  - Master the foundational building blocks of Deep Learning models.
  - Learn techniques for improving model performance and preventing overfitting.
  - Understand and implement transfer learning for image recognition tasks.
  - Explore practical applications of transfer learning using pre-trained Neural Networks.

4.4 Module Flow

Session	Module Flow
SESSION 1 (2.5 HOURS)	**<u>Feedforward Neural Networks – Architecture and Limitations</u>** **Description:** Study the structure and flow of data in feedforward neural networks. Understand their components, such as input layers, hidden layers, weights, biases, and activation functions. Discuss their limitations in handling spatial and temporal data, motivating the need for advanced architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Practical examples will include simple classification tasks. **Weightage of theory/hands-on for session:** • Theory: 100 mins • Hands-on: 50 min
SESSION 2 (4.5 HOURS)	**<u>Loss Functions and Backpropagation</u>** **Description:**

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	Learn how models measure error using loss functions, such as mean squared error (MSE) and cross-entropy loss. Explore how backpropagation computes gradients and adjusts weights to minimise loss. Hands-on exercises will involve visualising loss landscapes and implementing backpropagation in Python to solidify understanding. **Weightage of theory/hands-on for session:** • Theory: 180 mins • Hands-on: 90 min
SESSION 3 (2.5 HOURS)	**Regularisation Techniques** **Description:** Understand the causes and symptoms of overfitting in neural networks. Learn techniques like L1 and L2 regularisation, dropout, and early stopping to improve generalisation. Practical exercises will focus on applying these methods to stabilise training and avoid memorisation of noisy data. **Weightage of theory/hands-on for session:** • Theory: 110 mins • Hands-on: 40 min
SESSION 4 (3 HOURS)	**Hyperparameter Tuning** **Description:** Explore strategies for optimising hyperparameters, such as learning rates, batch sizes, and number of layers. Compare systematic approaches like grid search and random search. Practical exercises will involve tuning a CNN model and observing the effects of different configurations on training and validation performance. **Weightage of theory/hands-on for session:** • Theory: 120 min • Hands-on: 60 min
SESSION 5 (3.5 HOURS)	**Introduction to Transfer Learning and Pre-Trained Models** **Description:** Learn the concept of transfer learning, which reuses knowledge from pre-trained models to solve new tasks. Learn to access the Hugging Face Model Hub for pre-trained models. Study popular CNN architectures like ResNet, VGG, and EfficientNet, designed for large-scale datasets like ImageNet. Hands-on exercises will include loading a pre-trained model and adapting it for image classification tasks in a new domain. **Weightage of theory/hands-on for session:** • Theory: 140 min • Hands-on: 70 min
SESSION 6 (1 HOUR)	**Review and Assessment**

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	Description: The assessment will comprise a set of multiple-choice questions to reinforce the concepts covered in the module, as well as a practical component to test the participants on their ability to implement optimisation techniques.

4.5 **Who Should Attend**

- Software Developers seeking to enhance their skillset with Deep Learning expertise
- Data Scientists looking to expand their toolbox with powerful Deep Learning techniques
- Researchers and Engineers interested in applying Deep Learning to their specific domains
- Anyone with a passion for artificial intelligence and a desire to build intelligent systems

4.6 **Assessment**

4.6.1 Type of Assessment / Course Work

4.6.2 A) State the mode of assessment and provide a breakdown of the assessment duration in each mode of assessment.

Multiple-Choice Questions (Individual, open-book) – 20 min
Practical Component (Individual, open-book) – 40 min

4.6.3 B) Write up on why the assessment method is appropriate for the target audience/context/ application of the course and the assessment duration is reasonable.

The Multiple-Choice Questions are open-book as participants are not expected to memorise any information, but be able to search and test for the relevant information to arrive at the correct answers. Thus, ample time of more than 1 min per question is given.

The Practical Component is open-book as well, since data scientists and machine learning engineers and practitioners have the freedom to look up and reference any information they require to achieve their tasks.

4.6.4 C) Write up/ Assessment Plan on how the assessment can help to assess the knowledge gained from the training and demonstrate how the learning outcome can be met.

The Multiple-Choice portion is a straightforward and objective way to assess participants' grasp of theoretical knowledge covered during the module.

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The Practical Component assesses how well the participants are able to apply the skills that were covered in the module to a real-world example.

4.7 **Certification**

4.8 **Classroom Type**

4.9 **Special instructions for participants**

4.10 **Instructor's Bio and Head Shot (High Res Professional photo)**
Note: Please attach the photo in email as well.

4.11 **Supporting documents for verification process**

4.11.1 **Credentials mentioned in Bios.**

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5.0 Module OUTLINE

5.1 Module Title

Modern AI for Image and Natural Language Processing

5.2 Module Overview

This module explores how cutting-edge Artificial Intelligence (AI) models are applied to image and natural language processing tasks. Participants will gain an understanding of Convolutional Neural Networks (CNNs) and their role in image recognition and audio processing, alongside an introduction to Recurrent Neural Networks (RNNs) for sequential data. Advanced topics include Generative AI and Diffusion models, focusing on their practical applications in text processing and image generation. Real-world case studies and hands-on exercises using Python allow participants to implement and experiment with modern AI models for classification, clustering, and language understanding tasks. By the end, learners will be well-equipped to understand and leverage AI in image and text applications.

5.3 Module Objectives

- Learning Objectives
  - Describe the fundamental concepts of Convolutional Neural Networks (CNNs), including convolutional layers, filters, and pooling layers, and their role in tasks like image classification and object detection.
  - Implement CNNs using PyTorch or Tensorflow for tasks like image classification, leveraging common architectures like LeNet, VGG, and ResNet.
  - Describe the fundamentals of Recurrent Neural Networks (RNNs) and their applications in processing sequential data.
  - Gain an understanding of Generative AI and Duffusion Models for images and Natural Language Processing, with a specific focus on Stable Diffusion.

5.4 Module Flow

Session	Module Flow
SESSION 1 (3.5 HOURS)	**<u>Introduction to Convolutional Neural Networks (CNNs)</u>** **Description:** Learn the basic structure and operations of CNNs, including convolutions, filters, and pooling layers. Understand how CNNs extract hierarchical features from spatial data for tasks like image classification and object detection. Hands-on exercises include building a simple CNN using PyTorch or TensorFlow to classify basic image datasets. **Weightage of theory/hands-on for session:** • Theory: 140 mins

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	• Hands-on: 70 min
SESSION 2 (3.5 HOURS)	**<u>Advanced CNN Architectures</u>** **Description:** Explore cutting-edge CNN architectures, such as ResNet, DenseNet, and EfficientNet. Understand how architectural innovations, like skip connections and depth-wise separable convolutions, improve performance and scalability. Hands-on labs involve implementing a pre-trained CNN and fine-tuning it for a domain-specific task. **Weightage of theory/hands-on for session:** • Theory: 140 mins • Hands-on: 70 mins
SESSION 3 (2 HOURS)	**<u>Processing Audio Files Using an Image CNN</u>** **Description:** Learn how to process audio files by converting them into spectrograms, a visual representation of sound. Use CNNs to analyse these spectrograms for tasks like speech recognition and audio classification. Hands-on exercises include preprocessing audio data into spectrograms and training a CNN to classify audio events. **Weightage of theory/hands-on for session:** • Theory: 80 mins • Hands-on: 40 min
SESSION 4 (1.5 HOURS)	**<u>Introduction to Recurrent Neural Networks (RNNs)</u>** **Description:** Get an overview of RNNs and their role in sequence modelling tasks like text generation. Understand their limitations, such as vanishing gradients, and how they are addressed by advanced architectures like LSTMs and GRUs. Practical examples include a walkthrough of a simple RNN for text prediction. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 30 min
SESSION 5 (3 HOURS)	**<u>Introduction to Generative AI for Images and Natural Language Processing</u>** **Description:** Learn about Generative AI applications for images and natural language processing through the Stable Diffusion model, its architecture, and its ability to generate high-quality images from text prompts. Hands-on labs involve using pre-trained Stable Diffusion models to generate creative visual content and analyse the underlying diffusion process.

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	Weightage of theory/hands-on for session: • Theory: 60 min • Hands-on: 120 min
SESSION 6 (2.5 HOURS)	**Practical Applications of CNNs in Image Processing** **Description:** Explore real-world applications of CNNs, including image segmentation, object detection, and audio classification. Examine industry case studies like autonomous driving (image segmentation) and medical imaging (tumour detection). Practical exercises involve applying CNNs to real-world datasets in these domains. **Weightage of theory/hands-on for session:** • Theory: 50 min • Hands-on: 100 min
SESSION 5 (1 HOUR)	**Review and Assessment** **Description:** The assessment will comprise a set of multiple-choice questions to reinforce the concepts covered in the module, as well as a practical component to test the participants on their ability to create a neural network for image classification.

5.5 **Who Should Attend**

- Software Developers seeking to enhance their skillset with Deep Learning expertise
- Data Scientists looking to expand their toolbox with powerful Deep Learning techniques
- Researchers and Engineers interested in applying Deep Learning to their specific domains
- Anyone with a passion for artificial intelligence and a desire to build intelligent systems

5.6 **Assessment**

5.6.1 Type of Assessment / Course Work

5.6.2 A) State the mode of assessment and provide a breakdown of the assessment duration in each mode of assessment.

Multiple-Choice Questions (Individual, open-book) – 20 min

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Practical Component (Individual, open-book) – 40 min

5.6.3 B) Write up on why the assessment method is appropriate for the target audience/context/ application of the course and the assessment duration is reasonable.

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The Practical Component is open-book as well, since data scientists and machine learning engineers and practitioners have the freedom to look up and reference any information they require to achieve their tasks.

5.6.4 C) Write up/ Assessment Plan on how the assessment can help to assess the knowledge gained from the training and demonstrate how the learning outcome can be met.

The Multiple-Choice portion is a straightforward and objective way to assess participants' grasp of theoretical knowledge covered during the module.
The Practical Component assesses how well the participants are able to apply the skills that were covered in the module to a real-world example.

5.7 **Certification**

5.8 **Classroom Type**

5.9 **Special instructions for participants**

5.10 **Instructor's Bio and Head Shot (High Res Professional photo)**
Note: Please attach the photo in email as well.

5.11 **Supporting documents for verification process**

5.11.1 **Credentials mentioned in Bios.**

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6.0 Module OUTLINE

6.1 Module Title

Transformers, Large Language Models and Multimodal Learning

6.2 Module Overview

This module focuses on the transformative technologies driving modern Artificial Intelligence (AI) advancements. Participants will explore the architecture and applications of Transformers and Large Language Models (LLMs), including their use in tasks like text summarisation and question answering. Multimodal learning, which integrates data from multiple sources such as text and images, is also introduced. The module incorporates practical exercises on no-code platforms, enabling participants to experiment with pre-trained models for real-world tasks. Ethical considerations and AI safety are discussed to ensure responsible use of these technologies. By the end of this module, participants will have a comprehensive understanding of advanced AI models and their capabilities.

6.3 Module Objectives

- Learning Objectives
  - Describe the architecture and working principles of Transformers, including the attention mechanism and encoder-decoder architecture.
  - Describe the capabilities of Large Language Models (LLMs) and their significance in natural language understanding tasks.
  - Describe how multimodal models are able to simultaneously take inputs of varying formats and explore the capabilities of some of the commercially available multimodal models.
  - Address AI safety and ethical considerations for LLM deployment.

6.4 Module Flow

Session	Module Flow
SESSION 1 (3 HOURS)	**<u>Introduction to Transformers</u>** **Description:** Explore the general architecture of Transformers, with a focus on specific mechanisms such as self-attention, multi-head attention, encoder-decoder architectures and positional encoding. Understand why Transformers surpass RNNs in handling long-term dependencies and parallel processing at scale. Examine their role in advancing NLP, computer vision, and multimodal tasks. **Weightage of theory/hands-on for session:** • Theory: 120 mins • Hands-on: 60 min
SESSION 2	**<u>Large Language Models (LLMs)</u>**

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(5 HOURS)	**Description:** Understand the capabilities of Large Language Models (LLMs) and their significance in natural language understanding tasks. Explore pre-trained LLM architectures such as BERT and GPT-3, understanding their architectures and real-world applications. Coverage of cross-attention mechanisms with a case study on Stable Diffusion. Explore pre-training and fine-tuning paradigms, the use of masked language models, and autoregressive models. Study real-world applications of LLMs such as summarisation, chatbot development, and question answering. Discuss limitations, including hallucinations and biases. Discussion of empirical observations and phenomena like emergence and scaling laws which offer guidance to LLM practitioners in their design of LLM-based solutions. **Weightage of theory/hands-on for session:** • Theory: 100 mins • Hands-on: 200 min
SESSION 3 (3 HOURS)	**<u>Introduction to Fine-tuning LLMs</u>** **Description:** Hands-on session to fine-tune an LLM for domain-specific tasks using Hugging Face. Learn data preparation, tokenisation, and training strategies. Troubleshoot common issues such as overfitting and inadequate generalisation. Evaluate performance using MMLU, perplexity, BLEU, and F1-score metrics. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 120 mins
SESSION 4 (2.5 HOURS)	**<u>Multimodal Learning Overview</u>** **Description:** Explore multimodal models like CLIP and DALL-E that integrate text, images, and other modalities. Learn how these models enhance understanding and generation across diverse data types. Discuss applications in cross-modal retrieval, video captioning, and content creation. Hands-on exercise in Implementing a multimodal model using CLIP or a similar framework. Preprocess text and image data, train the model, and test its performance on cross-modal tasks like text-to-image retrieval. Highlight the challenges and advantages of working with multimodal datasets. **Weightage of theory/hands-on for session:** • Theory: 50 mins • Hands-on: 100 mins
SESSION 5 (2.5 HOURS)	**<u>AI Safety and Ethical Consideration</u>** **Description:** Discuss ethical concerns surrounding LLMs, such as biases in training data, potential for misinformation, and societal impact. Highlight strategies for ensuring

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	safe deployment, including monitoring for harmful outputs, transparency in AI development, and compliance with ethical guidelines. Explore case studies like GPT and its implementation challenges in customer service. **Weightage of theory/hands-on for session:** • Theory: 100 mins • Hands-on: 50 mins
SESSION 6 (1 HOUR)	**Review and Assessment** **Description:** The assessment will comprise a set of multiple-choice questions and practical component to reinforce the concepts covered in the module.

6.5 **Who Should Attend**

- Software Developers seeking to enhance their skillset with Deep Learning expertise
- Data Scientists looking to expand their toolbox with powerful Deep Learning techniques
- Researchers and Engineers interested in applying Deep Learning to their specific domains
- Anyone with a passion for artificial intelligence and a desire to build intelligent systems

6.6 **Assessment**

6.6.1 Type of Assessment / Course Work

6.6.2 A) State the mode of assessment and provide a breakdown of the assessment duration in each mode of assessment.

1. Multiple-Choice Questions (Individual, open-book) – 20 min
2. Practical Component (Individual, open-book) – 40 min

6.6.3 B) Write up on why the assessment method is appropriate for the target audience/context/ application of the course and the assessment duration is reasonable.

The Multiple-Choice Questions are open-book as participants are not expected to memorise any information, but be able to search and test for the relevant information to arrive at the correct answers. Thus, ample time of more than 1 min per question is given.

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The Practical Component is open-book as well, since data scientists and machine learning engineers and practitioners have the freedom to look up and reference any information they require to achieve their tasks.

6.6.4 C) Write up/ Assessment Plan on how the assessment can help to assess the knowledge gained from the training and demonstrate how the learning outcome can be met.

The Multiple-Choice portion is a straightforward and objective way to assess participants' grasp of theoretical knowledge covered during the module.

The Practical Component assesses how well the participants are able to apply the skills that were covered in the module to a real-world example.

6.7 **Certification**

6.8 **Classroom Type**

6.9 **Special instructions for participants**

6.10 **Instructor's Bio and Head Shot (High Res Professional photo)**

Note: Please attach the photo in email as well.

6.11 **Supporting documents for verification process**

6.11.1 **Credentials mentioned in Bios.**

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7.0 Module OUTLINE

7.1 Module Title

Deep Learning in Production

7.2 Module Overview

This module focuses on deploying Deep Learning models in real-world environments, with an emphasis on production-ready workflows, both on the cloud and on-premise/locally. Participants will learn to implement Retrieval-Augmented Generation (RAG), a technique that integrates external data retrieval to enhance Large Language Model (LLM) performance. Practical exercises cover tools like the LangChain framework, vector databases, and MLOps platforms to monitor and optimise deployed systems. The module also explores challenges such as scalability, reproducibility, and latency, with hands-on activities to address these issues. By the end of the module, learners will understand how to deploy, manage, and scale Deep Learning models effectively for business applications.

7.3 Module Objectives

- Learning Objectives
  - Understand the Pipeline paradigm for deploying Deep Learning models.
  - Learn model optimisation techniques for increased performance, greater efficiency and reduced model storage
  - Learn to deploy Deep Learning solutions to the cloud and on-premise
  - Implement Retrieval-Augmented Generation (RAG) for enhancing LLM performance with private data
  - Describe and evaluate the differences between the different offerings available for deep learning deployment. Compare and contrast between different frameworks, cloud platforms and options for local deployment

7.4 Module Flow

Session	Module Flow
SESSION 1 (1 HOURS)	**<u>Introduction to Production Pipelines</u>** **Description:** Learn about the lifecycle of deploying Deep Learning models, from data preprocessing to inference. Explore challenges in production environments, such as scalability, latency, and reliability. Discuss how the deployment pipeline integrates with larger enterprise systems. **Weightage of theory/hands-on for session:** • Theory: 20mins • Hands-on 40mins

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SESSION 2 (1 HOURS)	**Model Optimisation Techniques** **Description:** Learn optimisation techniques like quantisation, pruning, and distillation to improve training, inference speed and memory efficiency. Hands-on Python exercises involve optimising a deep learning model for real-time production scenarios. **Weightage of theory/hands-on for session:** • Theory: 20 mins • Hands-on 40 mins
SESSION 3 (4 HOURS)	**Deploying Models to the Cloud** **Description:** Hand-on experience in deploying a trained model on a cluster using tools and frameworks like Docker and Kubernetes, to various cloud providers like Amazon Web Service and Google Cloud. Set up dashboards and alerts to address MLOps concerns like monitoring model performance and drift in production. Learn how to use cloud logging tools and feedback loops to improve model reliability. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 180 min
SESSION 4 (4 HOURS)	**Deploying Models On-premise** **Description:** Introduction to the mainstream Linux/Nvidia-based server setup for LLM deployment, before contrasting with an Apple/MacOS-based alternative. Configure and run LLMs on a single local M-chip Mac server using Apple's MLX framework and PyTorch. Hands on experience in setting up open source LLMs on local multi-server Mac clusters for distributed inference. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 180 min
SESSION 5 (6 HOURS)	**Implementing Retrieval-Augmented Generation (RAG)** **Description:** Explore Retrieval-Augmented Generation (RAG), a widely adopted approach for building custom applications using Large Language Models (LLMs). RAG enhances LLM performance by integrating external knowledge, addressing challenges such as hallucination and limited context windows. Key use cases include chatbots, document summarisation, and personalised recommendations. This section covers LangChain, a framework for constructing RAG pipelines, through hands-on exercises like setting up retrieval systems with local databases. Participants will also learn LLamaIndex for organising and querying structured data, such as knowledge graphs, and gain an introduction to vector databases for

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	efficient similarity searches in large-scale datasets. Exploration of embeddings, highlighting their role in linking retrieval with generation and including practical exercises for generating embeddings using modern tools. **Weightage of theory/hands-on for session:** • Theory: 120 mins • Hands-on: 240 min
SESSION 6 (1 HOUR)	**Review and Assessment** **Description:** The assessment will comprise a set of multiple-choice questions to reinforce the concepts covered in the module, as well as a practical component to test the participants on their ability to deploy a trained model on a given Cloud platform.

7.5 **Who Should Attend**

- Software Developers seeking to enhance their skillset with Deep Learning expertise
- Data Scientists looking to expand their toolbox with powerful Deep Learning techniques
- Researchers and Engineers interested in applying Deep Learning to their specific domains
- Anyone with a passion for artificial intelligence and a desire to build intelligent systems

7.6 **Assessment**

7.6.1 Type of Assessment / Course Work

7.6.2 A) State the mode of assessment and provide a breakdown of the assessment duration in each mode of assessment.

1. Multiple-Choice Questions (Individual, open-book) – 20 min
2. Practical Component (Individual, open-book) – 40 min

7.6.3 B) Write up on why the assessment method is appropriate for the target audience/context/ application of the course and the assessment duration is reasonable.

The Multiple-Choice Questions are open-book as participants are not expected to memorise any information, but be able to search and test for the relevant information to arrive at the correct answers. Thus, ample time of more than 1 min per question is given.

The Practical Component is open-book as well, since data scientists and machine

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learning engineers and practitioners have the freedom to look up and reference any information they require to achieve their tasks.

7.6.4 C) Write up/ Assessment Plan on how the assessment can help to assess the knowledge gained from the training and demonstrate how the learning outcome can be met.

The Multiple-Choice portion is a straightforward and objective way to assess participants' grasp of theoretical knowledge covered during the module.
The Practical Component assesses how well the participants are able to apply the skills that were covered in the module to a real-world example.

7.7 **Certification**

7.8 **Classroom Type**

7.9 **Special instructions for participants**

7.10 **Instructor's Bio and Head Shot (High Res Professional photo)**
Note: Please attach the photo in email as well.

7.11 **Supporting documents for verification process**

7.11.1 Credentials mentioned in Bios.

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8.0 **Module OUTLINE**

8.1 **Module Title**

Tuning Deep Learning and Generative AI for Custom Applications

8.2 **Module Overview**

This module provides hands-on experience in fine-tuning Large Language Models (LLMs) to meet specific business requirements. Participants will learn techniques such as prompt engineering, parameter-efficient fine-tuning (PEFT), and reinforcement learning from human feedback (RLHF). The module also covers Deep Learning architectures and their role in pushing the boundaries of model scalability, efficiency and adaptability. Practical sessions allow learners to experiment with tuning LLMs for tasks like content generation and sentiment analysis. By the end of this module, participants will have the skills to customise and optimise LLMs for targeted applications, ensuring practical and flexible solutions.

8.3 **Module Objectives**

- Learning Objectives
  - Understand foundational and advanced techniques for fine-tuning large language models (LLMs).
  - Learn prompt engineering for efficient interaction with LLMs.
  - Explore parameter-efficient fine-tuning (PEFT), full fine-tuning, and reinforcement learning from human feedback (RLHF).
  - Finetune a deep learning model, from preparation and loading of the data to evaluation and analysis of the finetuned model, with insights drawn from real-world implementations and case studies

8.4 **Module Flow**

Session	**Module Flow**
SESSION 1 (2 HOUR)	**<u>Introduction to Fine-Tuning LLMs</u>** **Description:** Explore the need for fine-tuning LLMs for specific domains or tasks. Discuss pre-training versus fine-tuning, highlighting their differences and complementary roles. Introduce common scenarios, such as domain-specific customisation, improving accuracy, or aligning model outputs with organisational goals. Provide an overview of tools like Hugging Face for LLM fine-tuning. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 60 mins
SESSION 2 (3 HOURS)	**<u>Prompt Engineering</u>** **Description:**

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	Learn prompt design techniques to maximise the utility of LLMs without additional fine-tuning. Cover few-shot, zero-shot, and chain-of-thought prompting with practical examples. Highlight the importance of clear, structured prompts for steering model behaviour. Hands-on exercises include designing prompts for summarisation, creative writing, and sentiment analysis tasks. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 120 min
SESSION 3 (5 HOURS)	**<u>Parameter-Efficient Fine-Tuning (PEFT)</u>** **Description:** Implement PEFT techniques like LoRA (Low-Rank Adaptation), QLoRA and adapters that allow fine-tuning of smaller subsets of parameters, making training faster and less resource-intensive. Understand the Prompt Tuning technique as different approach to PEFT. Learn their applications in industry, especially when computational resources are limited. Hands-on sessions involve applying PEFT on an open-source LLM for domain-specific tasks. **Weightage of theory/hands-on for session:** • Theory: 120 mins • Hands-on: 180 min
SESSION 4 (2 HOURS)	**<u>Full Fine-Tuning of LLMs</u>** **Description:** Explore the process of fully fine-tuning LLMs. Cover the preparation of large datasets, tokenisation, optimiser settings, and distributed training strategies. Discuss challenges, including overfitting, data imbalance, and training cost. Hands-on exercises include preparing and fine-tuning a smaller model (like GPT-2) on a custom dataset. Compare Full Fine-Tuning to PEFT and Prompt Engineering and understand them as different options in the trade-off between model performance and training resource/time efficiency **Weightage of theory/hands-on for session:** • Theory: 30 mins • Hands-on: 90 mins
SESSION 5 (2 HOURS)	**<u>Reinforcement Learning from Human Feedback (RLHF)</u>** **Description:** Understand RLHF as a technique to align LLM behaviour with human preferences. Explore how human feedback is integrated into training using reward models. Case studies, such as OpenAI's GPT fine-tuning with RLHF, are covered. Hands-on sessions involve simulating RLHF workflows using smaller models and synthetic feedback **Weightage of theory/hands-on for session:** • Theory: 30 mins

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	• Hands-on: 90 mins
SESSION 6 (2 HOURS)	**Case Studies and Advanced Applications** **Description:** Examine real-world applications of fine-tuned LLMs across industries. Case studies include healthcare, legal document analysis, and creative industries. Reflect on the integration of techniques like PEFT, RLHF, and model quantisation for task-specific optimization. Hands-on exercises involve evaluating fine-tuned models for deployment readiness. **Weightage of theory/hands-on for session:** • Theory: 60 mins • Hands-on: 60 mins
SESSION 7 (1 HOUR)	**Review and Assessment** **Description:** The assessment will comprise a set of multiple-choice questions and practical component to reinforce the concepts covered in the module.

8.5 **Who Should Attend**

- Aspiring web developers seeking to develop their own modern web application
- Anyone with an interest in learning the fundamentals in web programming and developing web applications
- Professionals (IT and non-IT) looking to gain hands-on experience in developing their first web application

8.6 **Assessment**

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